

Laboratory session

DC Machines-DC Motors

Theoretical Fundamentals.

Operating Principle of DC Motors

The basic principle of a dc motor is the creation of a rotating magnet inside the mobile part of the motor, the rotor. This is accomplished by a device called the commutator which is found on all dc machines. The commutator produces the alternating currents necessary for the creation of the rotating magnet from dc power provided by an external source. Figure 1 illustrates a typical dc motor rotor with its main parts. This figure shows that the electrical contact between the segments of the commutator and the external dc source is made through brushes. Note that the rotor of a dc motor is also referred to as the armature.

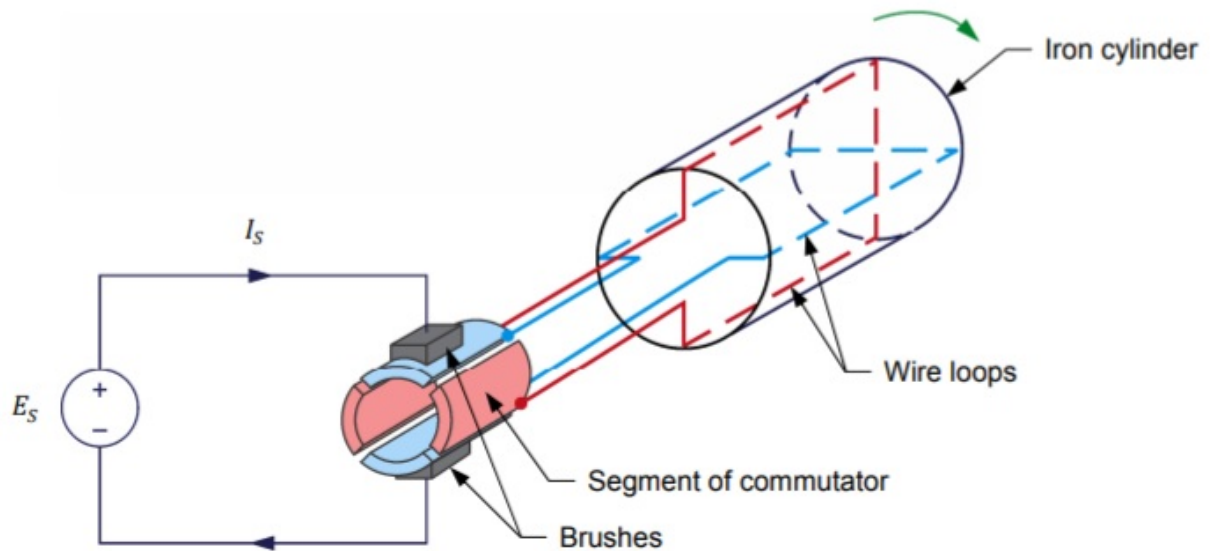


Figure 1. The Main Parts of a DC Motor Rotor (Armature)

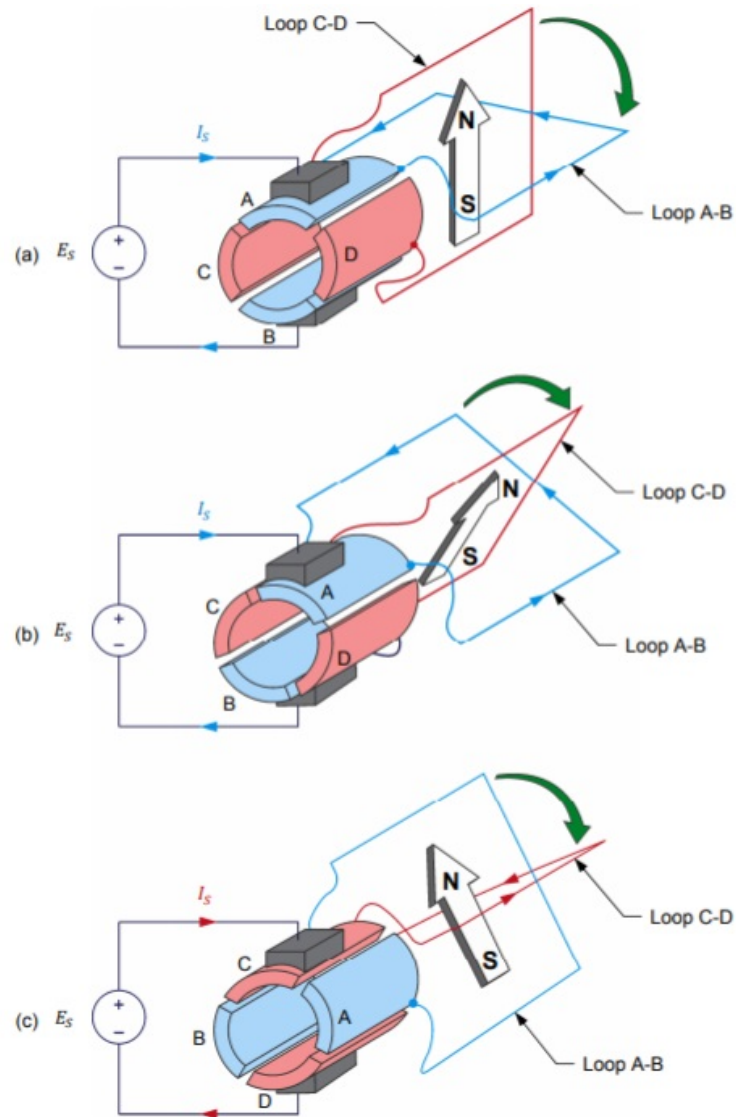


Figure 2. Operation of the Commutator.

In Figure 2 (a), the brushes make contact with segments A and B of the commutator and current flows in wire loop A-B. No current flows in the other wire loop (C-D). This creates an electromagnet A-B with north and south poles as shown in Figure 2 (a). If the rotor is turned clockwise a little as shown in Figure 2 (b), current still flows in wire loop A-B and the magnetic north and south poles rotate clockwise. As the rotor continues to rotate clockwise, a time comes where a commutation occurs, i.e. the brushes make contact with segments C and D instead of segments A and B, as shown in Figure 2(c). As a result, current now flows in wire loop C-D instead of flowing in wire loop A-B. This creates an electromagnet C-D with north and south poles as shown in Figure 2(c). By comparing Figures 2 (b) and (c), you can see that the magnetic north and south poles rotate 90° counterclockwise at the commutation. As the rotor continues to rotate clockwise, the same phenomenon repeats every 90° angle of rotation as shown in Figures 3 (a) to (c).

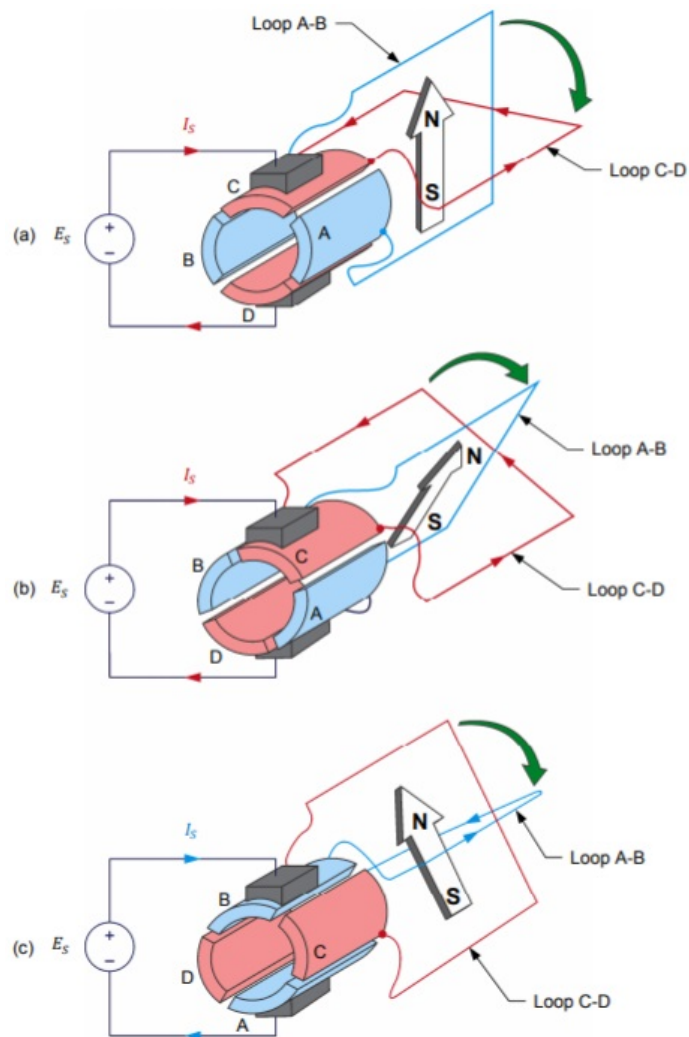


Figure 3. Operation of the Commutator (continued).

In brief, as the rotor turns, the north and south poles of the electromagnet go back and forth (oscillate) over a 90° angle as shown in Figure 4. In other words, the north and south poles are stationary, i.e. they do not rotate as the rotor turns. This is equivalent to having an electromagnet in the rotor that rotates at the same speed as the rotor but in the opposite direction. The higher the number of segments on the commutator, the lower the angle of rotation between each commutation, and the lower the angle over which the north and south poles oscillate. For example, the north and south poles would oscillate over an angle of only 11.25° if the commutator in Figures 2-2 to 2-4 were having 32 segments.

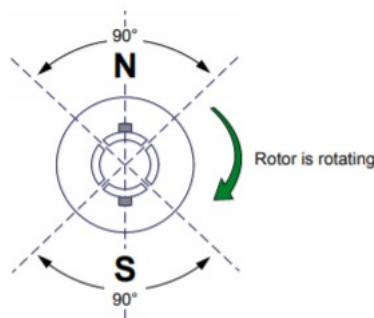


Figure 4. The North and South Poles Oscillate Around a Fixed Position.

If this rotor is placed next to a fixed permanent magnet stator as shown in Figure 2-5, the magnetic poles of opposite polarity attract each other (in order to align) and the rotor starts to turn. After the rotor has turned of a certain angle, a commutation occurs and the north and south poles of the electromagnet go back. Once again, the magnetic poles of opposite polarity attract each other, and the rotor continues to rotate in the same direction so as to align the magnetic poles of opposite polarity. However, another commutation occurs a little after and the north and south poles of the electromagnet go back once again. This cycle repeats over and over. The force that results from the interaction of the two magnetic fields always acts in the same direction, and the rotor turns continually. Thus, a converter of electrical-to-mechanical energy, i.e. an electric motor, has been achieved. The direction of rotation depends on the polarity of the voltage applied to the brushes of the rotor.

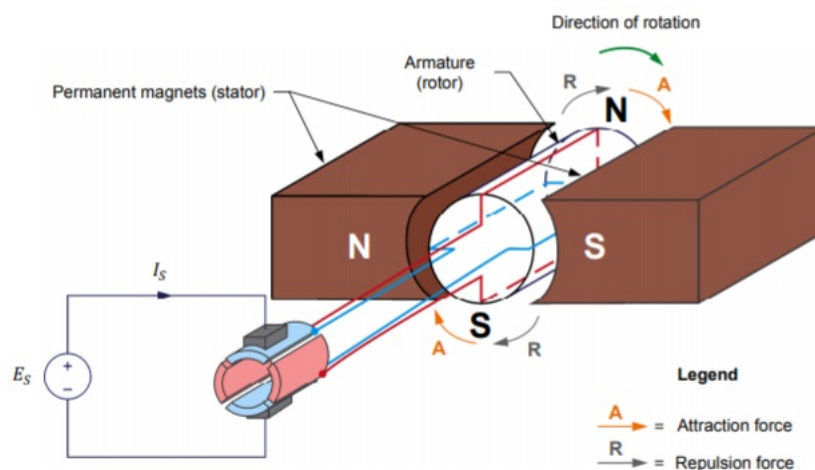


Figure 5. Rotation Resulting from Interaction of Magnetic Fields in the Stator and the Rotor.

The Separately-Excited DC Motor. Basic Operations.

EXERCISE OBJECTIVE

When you have completed this exercise, you will be able to demonstrate the main operating characteristics of a separately-excited dc motor using the DC Motor /Generator module.

DISCUSSION

Previously, you saw that a dc motor is made up basically of a fixed magnet (stator) and a rotating magnet (rotor). Many dc motors use an electromagnet for the stator, as illustrated in Figure 6.

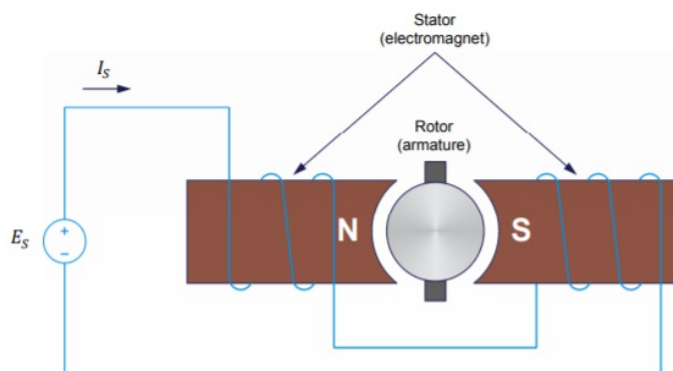


Figure 6. Simplified DC Motor with an Electromagnet as Stator.

When power for the stator electromagnet is supplied by a separate dc source, either fixed or variable, the motor is known as a separately-excited dc motor. Sometimes the term independent-field dc motor is also used. The current flowing in the stator electromagnet is often called field current because it is used to create a fixed magnetic field. The electrical and mechanical behaviour of the dc motor can be understood by examining its simplified equivalent electric circuit shown in Figure 7.

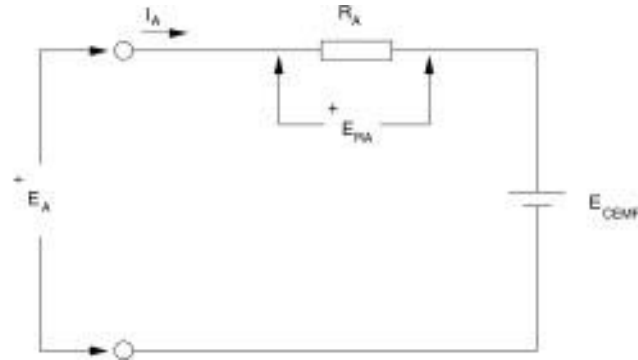


Figure 7. Simplified Equivalent Circuit of a DC Motor.

In the circuit, E_A is the voltage applied to the motor brushes, I_A is the current flowing through the brushes, and R_A is the resistance between the brushes. Note that E_A , I_A , and R_A are usually referred to as the armature voltage, current, and resistance, respectively. E_{RA} is the voltage drop across the armature resistor. When the motor turns, an induced voltage E_{CEMF} proportional to the speed of the motor is produced. This induced voltage is represented by a dc source in the simplified equivalent circuit of Figure 7. The motor also develops a torque T proportional to the armature current I_A flowing in the motor. The motor behaviour is based on the two equations given below. The first relates motor speed n and the induced voltage E_{CEMF} , and the second relates the motor torque T and the armature current I_A .

$$n = K_1 \times E_{CEMF} \quad \text{and} \quad T = K_2 \times I_A,$$

where K_1 is a constant expressed in units of r/min/V, K_2 is a constant expressed in units of N*m/A.

When a voltage E_A is applied to the armature of a dc motor with no mechanical load, the armature current I_A flowing in the equivalent circuit of Figure 7 is constant and has a very low value. As a result, the voltage drop E_{RA} across the armature resistor is so low that it can be neglected, and E_{CEMF} can be considered to be equal to the armature voltage E_A . Therefore, the relationship between the motor speed n and the armature voltage E_A is a straight line because E_{CEMF} is proportional to the motor speed n . This linear relationship is illustrated in Figure 8, and the slope of the straight line equals constant K_1 .

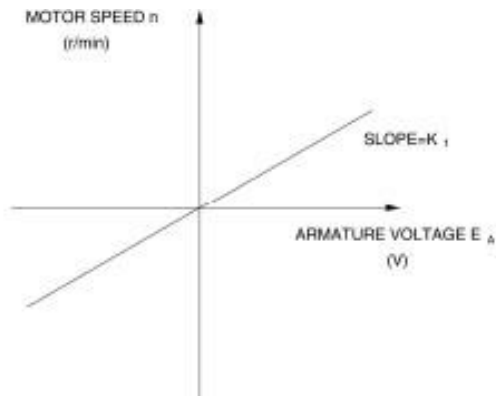


Figure 8. Linear Relationship between the Motor Speed and the Armature Voltage.

Since the relationship between voltage E_A and speed n is linear, a dc motor can be considered to be a linear voltage-to-speed converter as shown in Figure 9.

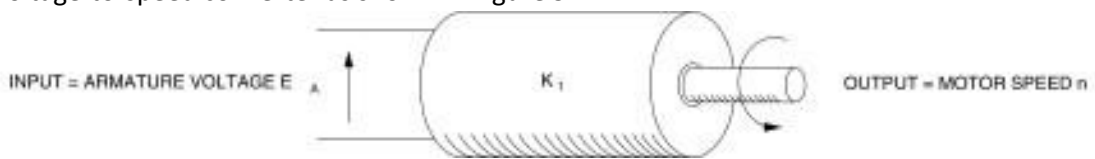


Figure 9. DC Motor as a Voltage-to-Speed Converter.

The same type of relationship exists between the motor torque T and the armature current I_A , so that a dc motor can also be considered as a linear current-to-torque converter. Figure 10 illustrates the linear relationship between the motor torque T and the armature current I_A . Constant K_2 is the slope of the line relating the two. In Figure 11, the linear current-to-torque converter is illustrated.

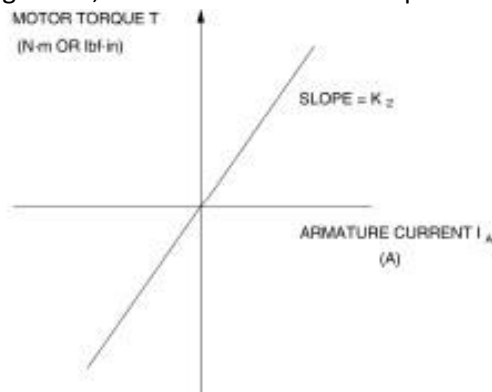


Figure 10. Linear Relationship between the Motor Torque and the Armature Current.



Figure 11. DC Motor as a Current-to-Torque Converter.

When the armature current I_A increases, the voltage drop E_{RA} ($R_A \times I_A$) across the armature resistor also increases and can no longer be neglected. As a result, the armature voltage E_A can no longer be considered to be equal to E_{CEMF} , but rather the sum of E_{CEMF} and E_{RA} as indicated in the following equation:

$$E_A = E_{CEMF} + E_{RA}$$

Therefore, when a fixed armature voltage E_A is applied to a dc motor, the voltage drop E_{RA} across the armature resistor increases as the armature current I_A increases, and thereby, causes E_{CEMF} to decrease. This also causes the motor speed n to decrease because it is proportional to E_{CEMF} . This is shown in Figure 12 which is a graph of the motor speed n versus the armature current I_A for a fixed armature voltage E_A .

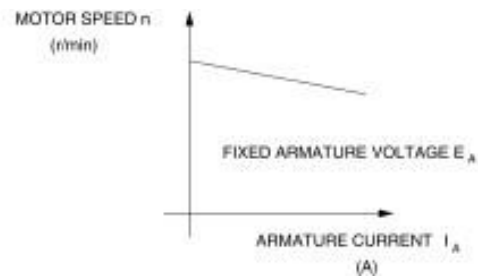


Figure 12. Motor Speed Drop as the Armature Current Increases (Fixed Armature Voltage E_A).

Procedure Summary

In the **first part** of the exercise, you will set up the equipment in the Workstation, connect the equipment as shown in Figure 2-14, and make the appropriate settings on the equipment.

In the **second part** of the exercise, you will measure the armature resistance R_A of the DC Motor / Generator. It is not possible to measure the armature resistance R_A directly with a conventional ohmmeter because the non-linear characteristic of the motor brushes causes incorrect results when I_A is too small. The general method used to determine the armature resistance R_A consists in connecting a dc power source to the motor armature and measuring the voltage required to produce nominal current flow in the armature windings. Power is not connected to the stator electromagnet to ensure that the motor does not turn, thus E_{CEMF} equals zero. The ratio of the armature voltage E_A to the armature current I_A yields the armature resistance R_A directly.

Note: The motor will not start to rotate because it is mechanically loaded.

In the **third part** of the exercise, you will measure data and plot a graph of the motor speed n versus the armature voltage E_A to demonstrate that the speed of the separately-excited dc motor is proportional to the armature voltage E_A under no-load conditions.

In the **fourth part** of the exercise, you will measure data and plot a graph of the motor torque T versus the armature current I_A to demonstrate that the torque of the separately-excited dc motor is proportional to the armature current I_A .

In the **fifth part** of the exercise, you will demonstrate that when the armature voltage E_A is set to a fixed value, the speed of the separately-excited dc motor decreases with increasing armature current or torque because of the increasing voltage drop across the armature resistor.

EQUIPMENT REQUIRED

Model	Description	Exercise 1
8134	EMS Work Station	1
8211	DC Motor/Generator	1
8311	Resistive Load	1

8942	Timing Belt	1
8960	Prime Mover and Dynamometer Module	1
8821	Power Supply	1
8951	Connection Leads	-
9063	Data Acquisition Module	1

PROCEDURE

This practice session required Risk Assessment Form to be produced before any action with equipment can be done Risk Assessment Form can be downloaded from CloudDeakin.



CAUTION!

High voltages are present in this laboratory exercise! Do not make or modify any banana jack connections with the power on unless otherwise specified!

Setting up the Equipment

1. Install the equipment required in the EMS workstation. Mechanically couple the prime mover / dynamometer module to the DC Motor / Generator using a timing belt.

Note: If you are performing the exercise using the EMS system, ensure that the brushes of the DC Motor / Generator are adjusted to the neutral point. To do so, connect an ac power source (terminals 4 and N of the Power Supply) to the armature of the DC Motor / Generator (terminals 1 and 2) through CURRENT INPUT I1 of the data acquisition module. Connect the shunt winding of the DC Motor / Generator (terminals 5 and 6) to VOLTAGE INPUT E1 of the data acquisition module. Turn the Power Supply on and set the voltage control knob so that an ac current (indicated by meter I line 1) equal to half the nominal value of the armature current flows in the armature of the DC Motor / Generator. Adjust the brush adjustment lever on the DC Motor / Generator so that the voltage across the shunt winding (indicated by meter E line 1) is minimum. Turn the Power Supply off, exit the Metering application, and disconnect all leads and cable.

2. On the Power Supply, make sure the main power switch is set to the O (off) position, and the voltage control knob is turned fully counter clockwise. Ensure the Power Supply is connected to a three-phase power source.

Note: If you are using the Four-Quadrant Dynamometer / Power Supply, Model 8960-B, connect its POWER INPUT to a wall receptacle.

3. Ensure that the data acquisition module is connected to a USB port of computer. Connect the POWER INPUT of the data acquisition module to the 24 V - AC output of the Power Supply.

Note: If you are using the Prime Mover / Dynamometer, Model 8960-1, connect its LOW POWER INPUT to the 24 V - AC output of the Power Supply.

On the Power Supply, set the 24 V - AC power switch to the I (on) position.

Note: If you are using the Four-Quadrant Dynamometer / Power Supply, Model 8960-B, turn it on by setting its POWER INPUT switch to the I (on) position. Press and hold the FUNCTION button 3 seconds to have uncorrected torque values on the display of the Four-Quadrant Dynamometer / Power Supply. The indication "NC" appears next to the function name on the display to indicate that the torque values are uncorrected.

4. Start the Metering application. In the metering window turn on I1, I2 and E1 windows.
5. Set up the separately-excited dc motor circuit shown in Figure 13. Leave the circuit open at points A and B shown in the figure.

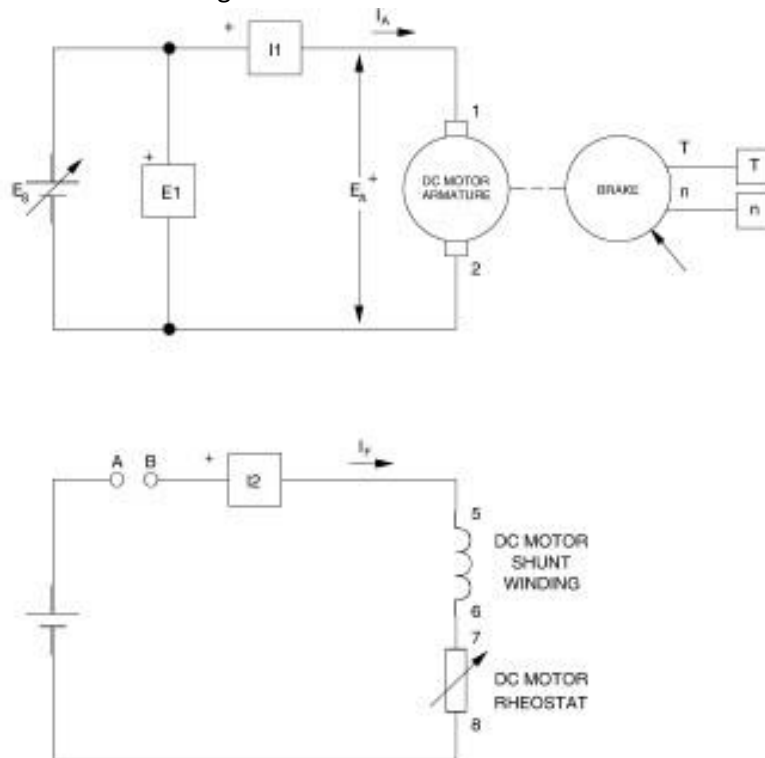


Figure 13. Separately-Excited DC Motor Coupled to a Brake.

6. Use the Function button to set the Four-Quadrant Dynamometer / Power Supply or the Prime Mover / Dynamometer to operate as a brake (2QT Brake), then set the torque control to maximum (fully CW position). This could also be done using the Prime Mover Dynamometer window in LVDAC software. If the dynamometer used has a computer controlled features.

Determining the Armature Resistance

7. Ask demonstrator to check you connections and turn the Power Supply on by setting its main power switch to the I (on) position, and set the voltage control knob so that the rated armature current flows in the DC Motor / Generator. The armature current is indicated by meter I arm. (IA) in the Metering window.

Note: The rating of any of the Lab-Volt machines is indicated in the lower left corner of the module front panel. If you are performing the exercise using LVSIM®-EMS, you can obtain the rating of any machine by leaving the mouse pointer on the rotor of the machine of interest. Pop-up help indicating the machine rating will appear after a few seconds.

Record the value of armature resistance R_A indicated by meter $R_A = E_A / I_A$.

8. Turn the voltage control knob fully counterclockwise and turn the Power Supply off. Interconnect points A and B shown in the circuit of Figure 13.

Motor Speed versus Armature Voltage

9. Turn the Power Supply on.
On the brake, set the torque control to minimum (fully CCW position).
On the DC Motor / Generator, set the FIELD RHEOSTAT so that the field current I_F indicated by meter I field (I_F) in the Metering window (I_2) is equal to the given value:

$$I_F = 210 \text{ mA}$$

10. In the Metering window for Prime Mover, make sure that the torque correction function of the Torque meter is enabled. The Torque meter now indicates the dc motor output torque. Record the armature voltage, armature current, field current, speed, and output torque in the Data Table. These parameters are indicated by meters E arm. (E_A), I arm. (I_A), I field (I_F), Torque, and Speed, respectively.

On the Power Supply, set the voltage control knob to 5%, 10%, 15% etc. up to 100% in order to increase the armature voltage E_A by steps. For each voltage setting, wait until the motor speed stabilizes, and then record the data in the Data Table.

11. When all data has been recorded, turn the voltage control knob fully counterclockwise and turn the Power Supply off. In the Data Table window, confirm that the data has been stored, entitle the data table as DT211, and save the data table.

12. In the Graph window, make the appropriate settings to obtain a graph of the dc motor speed n (obtained from the Speed meter) as a function of the armature voltage E_A [(obtained from meter E arm. (E_A)). Entitle the graph as G211, name the x-axis as Armature Voltage, name the y-axis as DC Motor Speed, and save the graph.

What kind of relationship exists between the armature voltage E_A and the dc motor speed n ? Does this graph confirm that the separately-excited dc motor is equivalent to a linear voltage-to-speed converter, with higher voltage producing greater speed?

13. Use the two end points to calculate the slope K_1 of the relationship obtained in graph G211. The values of these points are indicated in data table DT211. You can obtain this value in Excel program.

$$K_1 = \frac{n_2 - n_1}{E_2 - E_1} = \frac{-}{-} = \frac{\text{r/min}}{\text{V}}$$

In the Data Table window, clear the recorded data.

Motor Torque versus Armature Current

14. Turn the Power Supply on.
On the DC Motor / Generator, slightly readjust the FIELD RHEOSTAT so that the field current I_F indicated by meter I field (I_F) still equals the value given in Step 9 (if necessary).
On the Power Supply, set the voltage control knob so that the dc motor speed is **1500 r/min**. Note the value of the armature voltage E_A in the following blank space.

$$E_a = \underline{\hspace{2cm}} \text{ V, } n = 1500 \text{ r/min}$$

15. Record the dc motor output torque T , armature voltage E_A , armature current I_A , field current I_F , and speed n in the Data Table.

On the brake, set the torque control so that the torque indicated by the Torque meter in the Metering window increases by 0.1 Nm increments up to about 2.3 Nm. For each torque

setting, readjust the voltage control knob of the Power Supply so that the armature voltage E_A remains equal to the value recorded in the previous step, then record the data in the Data Table.

Note: The armature current may exceed the rated value while performing this manipulation. It is, therefore, suggested to complete the manipulation within a time interval of 5 minutes or less.

16. When all data has been recorded, set the torque control on the brake to minimum (fully CCW), turn the voltage control knob fully counterclockwise, and turn the Power Supply off. In the Data Table window, confirm that the data has been stored, entitle the data table as DT212, and save the data table.
17. In the Graph window, make the appropriate settings to obtain a graph of the dc motor torque (obtained from the Torque meter) as a function of the armature current I_A [obtained from meter I arm. (I_A)]. Entitle the graph as G212, name the x-axis as Armature Current, name the y-axis as DC Motor Torque, and print the graph.

What kind of relationship exists between the armature current I_A and the dc motor torque T as long as the armature current does not exceed the nominal value?

Does this graph confirm that the separately-excited dc motor is equivalent to a linear current-to-torque converter (when the armature current does not exceed the nominal value), with higher current producing greater torque?

Note: The torque versus current relationship is no longer linear when the armature current exceeds the nominal value because of a phenomenon called armature reaction. This phenomenon is described in the next unit of this manual.

18. Use the two end points of the linear portion of the relationship obtained in graph G212 to calculate the slope K_2 . The values of these points are indicated in data table DT212.

$$K_2 = \frac{T_2 - T_1}{I_2 - I_1} = \frac{\text{N} \cdot \text{m (lbf} \cdot \text{in)}}{\text{A}}$$

Speed Decrease versus Armature Current

19. Using the armature resistance R_A and the constant K_1 determined previously in this exercise, the armature voltage E_A measured in step 14, and the set of equations given below, determine the dc motor speed n for each of the three armature currents I_A given in Table 1 and fill the table values.

$$E_{RA} = I_A \times R_A$$

$$E_{CEMF} = E_A - E_{RA}$$

$$n = E_{CEMF} \times K_1$$

Local AC Power Network	Armature Current I_a	Armature Current I_a	Armature Current I_a
V_{ac}	A	A	A
240	0.5	1.0	1.5
E_{ra} (v)			

E_{cemf} (V)			
n (r/min)			

Table 1. DC Armature Currents and Voltages Calculations.

Based on your calculations, how should ECEMF and the dc motor speed n vary as the armature current is increased?

20. In the Graph window, make the appropriate settings to obtain a graph of the dc motor speed (obtained from the Speed meter) as a function of the armature current I_A [obtained from meter I arm. (I_A)], using the data recorded previously in the data table (DT212). Entitle the graph as G212-1, name the x-axis as Armature Current, name the y-axis as DC Motor Speed, and print the graph.

Does graph G212-1 confirm the prediction you made in the previous step about the variation of the dc motor speed as a function of the armature current I_A ?

Briefly explain what causes the dc motor speed to decrease when the armature voltage E_A is fixed and the armature current I_A increases.

21. In the Graph window, make the appropriate settings to obtain a graph of the dc motor speed (obtained from the Speed meter) as a function of the dc motor torque T (obtained from the Torque meter) using the data recorded previously in the data table (DT212). Entitle the graph as G212-2, name the x-axis as Separately-Excited DC Motor Torque, name the y-axis as Separately-Excited DC Motor Speed, and print the graph. This graph will be used in the next exercise of this unit.
22. Set the 24 V -AC power switch to the O (off) position, and remove all leads and cable.

The Separately-Excited DC Motor

REVIEW QUESTIONS

1. What kind of relationship exists between the speed and armature voltage of a separately-excited dc motor?
 - a. A linear relationship.
 - b. A parabolic relationship.
 - c. An exponential relationship.
 - d. The speed of the motor is independent of the applied voltage.
2. What kind of relationship exists between the torque and armature current of a separately-excited dc motor as long as the armature current does not exceed the nominal value?
 - a. A linear relationship.
 - b. A parabolic relationship.
 - c. An exponential relationship.
 - d. The motor torque is independent of the current.
3. Connecting a dc source to the armature of a dc motor that operates without field current and measuring the voltage that produces nominal current flow in the armature allows which parameter of the dc motor to be determined.
 - a. The nominal armature current.
 - b. The nominal armature voltage.
 - c. The armature resistance.
 - d. The resistance of the field winding.
4. Does the speed of a separately-excited dc motor increase or decrease when the armature current increases?
 - a. It increases.
 - b. It decreases.
 - c. It stays the same because speed is independent of motor current.
 - d. The speed will oscillate around the previous value.
5. The armature resistance R_A and constant K_1 of a dc motor are 0.5Ω and 5 r/min/V , respectively. A voltage of 200 V is applied to this motor. The no-load armature current is 2 A . At full load, the armature current increases to 50 A . What are the no-load and full-load speeds of the motor?
 - a. $n_{\text{NO LOAD}} = 1005 \text{ r/min}$, $n_{\text{FULL LOAD}} = 880 \text{ r/min}$
 - b. $n_{\text{NO LOAD}} = 995 \text{ r/min}$, $n_{\text{FULL LOAD}} = 875 \text{ r/min}$
 - c. $n_{\text{NO LOAD}} = 1000 \text{ r/min}$, $n_{\text{FULL LOAD}} = 875 \text{ r/min}$
 - d. The speeds cannot be calculated without constant K_2 .